



FINAL PROJECT PROPOSAL

Car Connect

PREPARED FOR

COM S 3190 - Construction of User Interfaces
Iowa State University Computer Science Department

PREPARED BY

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1. Introduction

We, Martin Kochadampally and Matthew Gathman, are from group JK_5. Martin is a sophomore, and Matthew is a junior, and both of us do not have any previous experience in web development. We are majoring in Computer Science, and this semester, we are taking COM S 3190, where we learn to develop User Interfaces. So far, we have learned to use HTML, CSS, Javascript, and React, and we plan to use React, in addition to Node.js and MySQL, to develop a website about cars for our final project.

2. Purpose of the Proposal

The purpose of this project is to create a visually engaging and responsive car showcasing website. Choosing the right car can be a difficult decision , and we want to help make that process easier by providing the user with a wide collection of cars with images, specifications and key details. Our website Car Connect is designed to help users explore the various models and help them to connect with their dream car.

3. Goals and Objectives

Our goals and objectives of CarConnect are:

- Develop a user-friendly website with an intuitive design.
- Provide detailed information about cars and allow the users to favorite and create reviews to help build user engagement.
- Demonstrate proficiency in Node.js, React, MySQL, and the REST API by creating a visually appealing website with a full stack.

4. Project Description

Core Features:

- User Auth: Login/Signup.
- Car Listings: Filter by make, year, or price.
- Favorites System: Save cars to user profiles (CRUD).
- Reviews: Rate and comment on listings (CRUD).
- Submission Portal: Users add new cars.

5. Project Path Selection

We are choosing **Option 2: Propose a new project** for our Final Project. While there is still room for improvement on our previous website, *CyExplore*, we believe starting a new project will give us more creative freedom and allow us to better showcase our new skills.

6. Feature Ownership and Responsibilities

Feature	Assigned To
Login & Signup	Matthew G (Frontend+Backend)
Home Page + Car Listing	Martin K (Frontend+Backend)
Review System	Matthew G (Frontend+Backend)
Favorite System	Martin K (Frontend+Backend)

7. Resources and Tools

- Frontend: React, Bootstrap.

- Backend: Node.js, Express, MySQL.
- Design: Figma (wireframes) and/or Excalidraw.
- Tools: GitLab, Postman (API testing), and the REST API.
- Responsibilities will be divided equally between Martin and Matthew. We will both commit around 5-10 hours per week to this project.

8. File Structure and Project Organization

frontend/

src/

assets/ # Car images, icons

components/ # Navbar, CarCard, ReviewForm

pages/ # Home, Login, Dashboard

backend/

Documents/ # Wireframes, SAD, demo video

The front will receive/send the data and type of request given by the user to or from the database.

9. Data Sources and Management

[

{

"id": 1,

"make": "Toyota",

"model": "Camry",

"year": 2023,

"price": 27500,

"color": "Midnight Black",

"image_url": "https://example.com/camry.jpg",

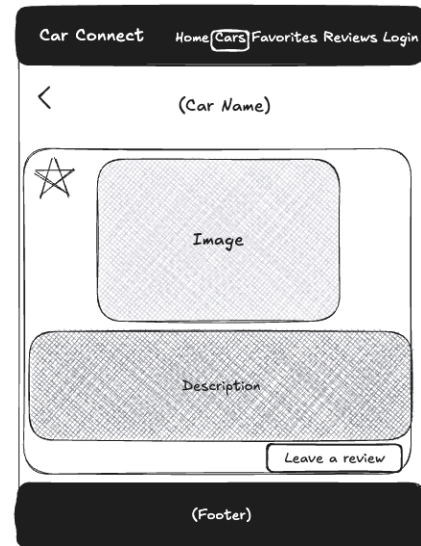
```
"specs": {  
  "engine": "2.5L 4-cylinder",  
  "mpg": "32 city / 41 highway",  
  "features": ["Apple CarPlay", "Heated Seats"]  
}  
}  
]
```

Our data will come in from User Input, and the sample json above could be something that we are looking for when using the cards for displaying the different cars. A CRUD flow could look like this: User submits review → POST /api/reviews → MySQL update → Display on the car details page.

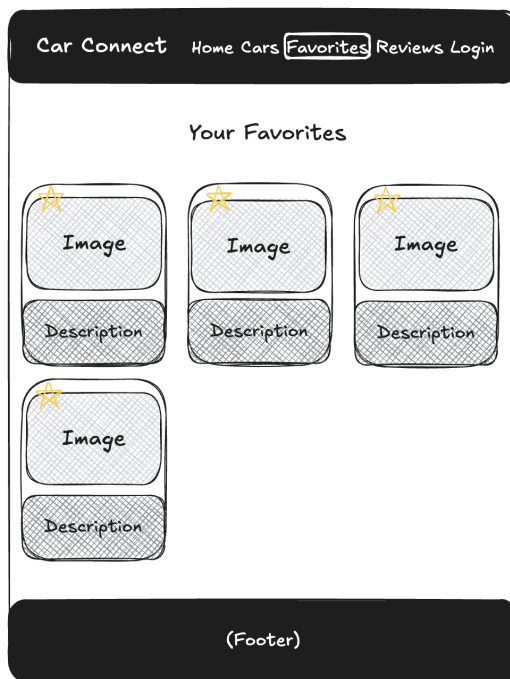
10. User Experience View

The image shows two side-by-side wireframe cards for a service named 'CarConnect'. Each card has a rounded rectangular shape with a light gray border. The left card is titled 'Log in' and contains two input fields labeled 'Username' and 'Password'. The right card is titled 'Sign Up' and also contains two input fields labeled 'Username' and 'Password'. Both cards have a light gray header bar at the top with the 'CarConnect' logo. The right card has a light gray bar at the bottom.

Log In and Sign up. The user will see the fields to input their username and password to either create an account or log in. The user will have the option to go from log in to sign up and after signing up it will automatically bring you to log in.



This is the Cars page. The images will be presented using cards. Each card will have an image and a description about the image. When you click on a card, it will expand with a more detailed description and you will also be able to leave a review with the leave a review button taking you to the reviews page.



This is the Favourites page. When you favourite a car, it will come on this page. This will look different for each user, based on what cars they decide to favourite.

CarConnect

Reviews

Car Image

Car Description

★★★★☆

Title of Car

Review Title

Review Description

Submit

Cancel

This is the review page where the user will be able to add stars, a title, and a description for the review. They will also be able to edit, and delete their own reviews. Users will also be able to see the original post that they are reviewing.

11. Final Comments

This is our proposal for CarConnect. Not only will this project give us a chance to showcase our skills, but it will also be an excellent resource for people who are already interested in cars and people who are not as educated on the topic.

We would like to learn more about implementing a full-stack into a website using React and Node.js.

Thank you for taking the time to consider our proposal. If you have any suggestions, we would love to hear them. Please reach out to us at:

martink5@iastate.edu or mattgath@iastate.edu